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OBJECTIVE

A Computer Science and Engineering graduate with internship experience in AI research and fintech, with a growing interest in Artificial Intelligence and Cybersecurity, particularly at their intersection. Currently contributing to research on multi-agent systems for secure microservice code generation, vulnerability detection, and remediation, while gaining industry experience working with data and AI models in the fintech domain.

Research Interests

- Large Language Models (*LLM Reasoning, Fine-tuning, Model Editing, RAG, Efficient Inference, Parameter-Efficient Learning*)
- AI Agents (*Agentic AI, Multi-Agent Systems, Agent Security, Guardrails, Tool-Augmented LLMs, Agent Evaluation*)
- AI for Cybersecurity (*Code Vulnerability Analysis, Intrusion Detection, Threat Detection, Security Automation, Malware Analysis*)

EXPERIENCE

Nagad Ltd

N-Risers Intern – Business Assurance Department

Banani, Dhaka, Bangladesh

August 2026 – Present

- Working with Scala and Python languages for data analysis, experimentation, and data-driven business applications.
- Applying AI models to automate manual processes and support operational tasks within the fintech domain.

Elite Research Lab LLC

Research Intern

Remote

June 2026 – Present

- Conducting research on secure and trustworthy AI systems, focusing on LLM and AI agent security.
- Contributing to structured reasoning workflows, literature review, experimental design, and technical documentation.

EDUCATION

North South University

Bachelor of Science in Computer Science and Engineering (BSCSE)

Dhaka, Bangladesh

February 2021 – September 2025

- CGPA: 3.04/4.00 (Equivalent to $\approx 84\%$)
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Advance Database, Machine Learning, Deep Learning.

ONGOING RESEARCH

A Secure Autonomous Agentic Framework for Microservice Code Generation with Automated Vulnerability Detection and Remediation

Elite Research Lab LLC (Remote Collaboration)

- Investigating secure autonomous AI agents for microservice code generation to improve AI-generated software security.
- Conducted literature review on agentic AI, secure code generation, and vulnerability remediation approaches.
- Developing an agentic framework integrating static security tools (e.g., SonarQube) to enable iterative vulnerability detection and code improvement.

THESIS WORK

BSc Thesis — An Explainable Approach for Network Intrusion Detection Using Machine Learning and Deep Neural Network

August 2024 – April 2025

Dept. of Electrical and Computer Engineering, North South University

This research tackles network security by building a robust intrusion detection system driven by a custom weighted ensemble classification model, while actively investigating the impact of SMOTE-generated synthetic data on class imbalance across varying dataset scales. To ensure these high-accuracy predictions are practical and transparent, the system integrates SHAP and LIME frameworks to clearly explain the reasoning behind every flagged threat.

- Developed an explainable, weighted ensemble classification model—combining Decision Tree and Random Forest classifiers—utilizing SHAP and LIME to ensure transparent and interpretable threat detection.
- Evaluated the impact of synthetic data on model performance by generating and analyzing three different configurations of the CIC-IDS2017 dataset using SMOTE.
- Achieved 97.26% accuracy and a remarkably low false positive rate of 0.0048 in classifying, multi-class network intrusions.
- Links — [Poster](#) [Report](#) [Codes](#)

PROJECTS

CyberSec-Reasoner – A Three-Stage Post-Trained Reasoning LLM for Vulnerability Analysis [GitHub](#)

PyTorch, Hugging Face Transformers, TRL, Hugging Face PEFT, LoRA, QLoRA, GRPO, Qwen 3.5, Wandb.

- Engineered a domain-specialized LLM based on Qwen 3.5 (4B) to perform Chain-of-Thought reasoning for CVE-to-CWE mapping.
- Architected a three-stage pipeline combining Knowledge SFT, Cold Start SFT, and GRPO-based reinforcement learning.
- Developed a weighted hybrid reward system ($\text{Total_Reward} = 0.5 \cdot R_{cwe} + 0.3 \cdot R_{format} + 0.2 \cdot R_{keyword}$) for classification accuracy and semantic grounding, validating the model against a 1,000-sample CVE/CWE dataset with rule-based metrics.

Multi-lingual YouTube Comment Sentiment & Type Analysis [GitHub](#)

Scikit-learn, PyTorch, Hugging Face Transformers, FastAPI, PostgreSQL, Postman.

- Developed a multilingual NLP pipeline using XLM-RoBERTa for sentiment and comment-type classification.
- Automated dataset collection and labeling using the YouTube API and Qwen2.5-14B.
- Trained and evaluated multiple models, achieving up to 76.68% sentiment and 76.05% type accuracy on test data.
- Built a FastAPI service with visualization tools for comment and video-level analysis.

SKILLS

Languages	Python, Scala, Java, C++, SQL, HTML/CSS
Library/Frameworks	Scikit-learn, TensorFlow, PyTorch, Transformers, LangChain, FastAPI
Tools & Platforms	Git & GitHub, LaTeX, Postman, Virtual Machine, MySQL, PostgreSQL, WanDB, Windows, Linux
Networking & Security	TCP/IP, UDP, HTTP/HTTPS, DNS, DHCP, SSH, OSI Model, CVE, CWE
Concepts	Machine Learning, Deep Learning, Large Language Models, Generative AI, Agentic AI, Object-Oriented Programming, Debugging, Prompt Engineering, Offensive & Defensive Security

ACTIVITIES

Participant, Google Cloud Gen AI Academy APAC Edition — *Hack2Skill × Google Cloud* February 2026 – Present

- Completed Cohorts 1 & 2, earning GenAI completion certifications — [Cohorts 1](#) [Cohorts 2](#)
- Currently attending Cohort 3, building LLM, AI agent, and GenAI development skills.

Participant, Aspire Leaders Program — *Aspire Institute* July 2026 – Present

- Selected for a global leadership development program.
- Developing leadership, communication, networking, and career skills.

Participant, ML Summer School 2026 — *Cohere Labs* July 2026 – August 2026

- Participated in 12 sessions, gaining practical exposure to modern ML, LLMs, AI agents, and ML research — [credentials](#)

CERTIFICATIONS

- Pre Security Learning Path** – TryHackMe – [Credentials](#) May 2026
- Associate AI Engineer for Developers** – DataCamp – [Credentials](#) Apr. 2026
- Machine Learning Specialization** – Coursera – [Credentials](#) Dec. 2024
- Google IT Automation with Python** – Coursera – [Credentials](#) Aug. 2024

ADDITIONAL DETAILS

Soft Skills: Problem Solving, Research-oriented, Team Collaboration, Time Management, Self-driven, Presentation Skills.
Languages: Bengali (Native), English (B2), Hindi (A2).

REFERENCES

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